

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks:50

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I VAIBHAVAK LAKSHMI K (192511310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

1. A network administrator has configured a workstation with the following settings:
- a. **IP Address:** 192.168.10.65

b. **Subnet Mask:** 255.255.255.192 (/26)

c. **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

ANSWER:

1. Identify the subnet to which the configured IP address belongs

Explanation:

The workstation is assigned the IP address **192.168.10.65** with the subnet mask **255.255.255.192 (/26)**. A /26 subnet divides the network into blocks of **64 IP addresses**. Since 192.168.10.65 falls between **192.168.10.64 and 192.168.10.127**, it belongs to the **192.168.10.64/26** subnet. Identifying the correct subnet helps determine whether devices can communicate within the same network

2. An organization has allocated the network **198.168.8.0/24** into several departments using the Subnet mask **198.168.8.0/26**. Network Engineer need additional IP addresses, which offers more a large number of unused addresses.

Task:

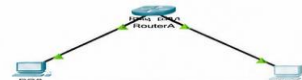
Analyse and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the total number of usable host addresses.
- Identifying the subnet with the most hosts.
- Debugging the addressing plan by locating subnets allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental required.

ANSWER:

RouterA#show ip route

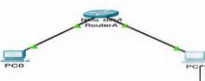
Device	IPv4 Address	Prefix Length
PC0	2001:db8::200:42:81:2c	/64
PC1	2001:db8:0:0:0:1::2	/64



RouterA# show ipv6 route

IPv6 Routing Table - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
I - IGRP, R - RIP, II - IS-IS, O - OSPF
Gateway of last resort is not set

C	2001:DB8::/64 is directly connected, Serial0/0/0
L	2001:DB8::1/328 is directly connected, Serial0/0/0
C	2001:DB8:1::/64 is directly connected, GigabitEthernet0/0
L	2001:DB8:1:1/128 is directly connected, GigabitEthernet0/0



Router A# show ipv6 route

IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
I - IGRP, R - RIP, II - IS-IS, O - OSPF
Gateway of last resort is not set

C	2001:DB8::/64 is directly connected, Serial0/0/0
L	2001:db8::1/128 is directly connected, Serial0/0/0

Expanded IPv6 Address

A: 2001:db8:

Compressed Address: 2001:db8:0000:0000:0000:0001:0000:0000

Expanded Address: 2001:0db8:0000:0000:0000:0001:0000:0000:0000:0000

Router B

Compressed Address: 2001:db8:0:0:0:1::2

Expanded Address: 2001:0db8:0000:0000:0000:0001:0000:0000:0000:0002

Network Prefix Calculation

Network prefix length is /64, the first four hextets represent the network portion.

Device A:

2001:db8:0:0::/64

Device B:

2001:db8:0:0::/64

Both devices belong to the same subnet.

Debugging Scenario

In debugging the ipv6 troubleshooting, the IPv6 address of PC1 was intentionally modified.

Modified IPv6 Address of PC1:

2001:db8:0:0::1:1:2:4

PC's Networks

PC0: 2001:db8::/64

PC1: 2001:db8::/64

Since network prefix portions are different, both devices belong to different subnets, causing communication failure.

2. Determine the Network Address, Broadcast Address and Valid Host Range

Explanation:

Once the subnet is identified, the network details can be calculated.

- Network Address = **192.168.10.64**
- Broadcast Address = **192.168.10.127**
- Valid Host Range = **192.168.10.65 – 192.168.10.126**

The network address identifies the subnet, the broadcast address is used to communicate with all hosts in the subnet, and the host range specifies the valid IP addresses that can be assigned to devices.

3. Verify whether the configured IP Address and Gateway are valid

Explanation:

The workstation IP address **192.168.10.65** is valid because it falls within the valid host range. However, the default gateway **192.168.10.1** belongs to a different subnet (**192.168.10.0/26**). Since the PC and gateway are not in the same subnet, communication fails. Both must belong to the same subnet for proper routing.

4. Identify the Configuration Errors

Explanation:

The main configuration error is that the workstation and the default gateway belong to different subnets. Although the IP address is correct, the gateway configuration is incorrect. This mismatch prevents the workstation from reaching other devices on the network.

Errors Found:

- Incorrect default gateway
- Gateway belongs to another subnet
- Communication failure due to subnet mismatch
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5. Optimize the Configuration

Explanation:

To solve the issue, assign both the workstation and the gateway addresses from the same subnet. A corrected configuration is:

- PC IP Address = **192.168.10.66**
- Subnet Mask = **255.255.255.192**
- Default Gateway = **192.168.10.65**

After correction, both devices belong to the **192.168.10.64/26** subnet, allowing successful communication.

6. Calculate the Total Number of Usable Host Addresses

Explanation:

A /26 subnet provides **6 host bits**.

- Total IP Addresses = $2^6 = \mathbf{64}$
- Reserved Addresses = 2
 - Network Address
 - Broadcast Address
- Usable Host Addresses = $\mathbf{64 - 2 = 62}$

Therefore, the corrected subnet supports **62 usable host addresses**.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation